

**The Sink Is a Flattening:  
Charge, Antigravity, and the Photon in Flowing Space**

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## Abstract

We extend the flowing-space framework in two geometric directions and join them into one chain.

(i) *The half-phase of the critical sheet*: the inversion fixed point of the envelope reflection  $w \mapsto e/w$  sits at  $|w| = \sqrt{e}$ , so  $\ln(\sqrt{e}) = 1/2$ ; the square root is the half-step operator of the multiplicative group, and in phase space the same half-step is the half-angle phase  $\exp(i\theta/2)$ , whose closure factor is  $-1$  — the explicit fermionic witness of spin  $1/2$ . (ii) *Charge as flow divergence*: positive charge is a source (positive divergence), negative charge a sink (negative divergence), mutually exclusive, flipped under  $\rho \rightarrow -\rho$ , neutral at zero. A time-varying source forces the space field to vary (CR1); a varying field activates the nuclear channel  $m dC$  of the four-force decomposition (CR2); and a sink — a converging flow — is geometrically a flattening: the inward contraction  $v_i \mapsto (1-\lambda)v_i + \lambda\bar{v}$  reduces the fluctuation energy  $Q_A$  by the square factor  $(1-\lambda)^2$  (CR9, exact discrete theorem). Joining the framework’s own  $\mu$ -bridge (one flattening gives  $\mu = 1$ ), the chain closes conditionally: accelerated charge  $\Rightarrow$  antigravity field  $\Rightarrow$  the photon (mass zero and neutral). As a device-level application, the self-flattening of circulating electrons shortens the approach to the working window  $\mu = 0.999$  from 132 to 85 steps and reaches it with halved external RMF drive. All statements are formalized in Lean 4 with zero `sorry`; the honest status is stated:  $\lambda$  (the sink contraction rate) and  $\delta, \varepsilon$  (the sink energy scale) are inputs, and the continuum 3D divergence-to-functional correspondence is not yet formalized.

*“Any road followed precisely to its end leads precisely nowhere. Climb the mountain just a little bit to be sure that it’s a mountain.”*

— Frank Herbert, *Dune* (1965)

## I. INTRODUCTION

*a. Four axioms in conflict.* Modern physics rests on four conceptual contradictions that are not technical difficulties but conflicts of axioms: the superposition of spacetime itself (quantum superposition is linear, while the general-relativistic background is dynamical [1]); the measurement problem (collapse is not in the Schrödinger equation); the black-hole information paradox (unitarity versus the horizon); and the vacuum-energy catastrophe

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(some  $10^{120}$  orders of magnitude). Standard responses keep the axioms and add entities — extra dimensions, new fields, new particles. This series of papers follows the opposite strategy: *change the primitive*. The first paper of the series [2, 3] showed that a single entity, *flow*, carries the kinematics of physics: from a small set of postulates — vector light speed, three anchored directions, and the counting of space-displacement strands — it derives mass, information, causality, the four forces, and the quantum relations, all as exact identities formalized in Lean 4 with zero **sorry**.

*b. The primitive is flow.* In this worldview space itself is a flowing medium — the same geometric structure as the “river model” of black holes [4], in which space flows inward through the horizon like water through a waterfall. A physical object is an *anchored pattern* of the flow: the photon is a pattern carried completely by the flow, while mass is the anchoring of a pattern against it. From the helix geometry of strands, the framework recovers the quantum relations that historically entered physics one by one — the Planck–Einstein quantization [5], the de Broglie wave [6], the Heisenberg uncertainty product [7] — now as consequences of one geometry rather than as postulates.

*c. Charge: the most familiar and least understood quantum number.* Among the quantum numbers, charge is special. It is conserved; it comes in integer multiples of  $e$ ; and its accelerations radiate photons — the one process by which the classical world routinely manufactures quanta. Yet no axiom of the standard model explains why these facts hold. In flowing space the question acquires a geometric answer: charge is the *divergence* of the space-displacement flow. A positive charge is a source from which displacement diverges; a negative charge is a sink into which it converges; the sign of the divergence is the sign of the charge. This paper’s first addition turns that statement into theorems, and follows where it leads: a sink is a converging flow, convergence is flattening, and a fully flattened flow is massless and neutral — the photon.

*d. The second addition: the half-phase geometry.* Where does the value  $1/2$  itself come from? Spin  $1/2$  is the simplest fermionic quantum number, yet in standard physics it is entered by hand — Dirac wrote it into the equation [8], Pauli read it off the exclusion principle [9], but nobody has derived it from geometry. The present work locates it: the Riemann–Hilbert envelope of the framework [10] carries the reflection  $s \mapsto 1 - s$ , whose fixed point is necessarily  $1/2$ ; in the logarithmic coordinate  $w = e^s$  that fixed point is  $\ln(\sqrt{e}) = 1/2$ , and the square root is recognized as the half-step operator of the multiplicative

group — in phase space, the half-angle phase  $\exp(i\theta/2)$  whose closure factor is  $-1$ .

*e. One fact, two faces.* The two additions are independent in content but share one theorem-level fact: *a quadratic quantity is reduced by a square factor*. For spin, the half-angle phase returns to itself with factor  $-1 = (-1)^2$ -free of any square: the half step of the variable  $\theta$  is the square root in logarithmic coordinates, and the fermionic witness is  $\exp(i\theta/2)$ . For charge, the fluctuation energy of a converging flow is reduced by  $(1 - \lambda)^2$  per contraction step; the sink is a self-flattening structure, and a sink at full strength ( $\lambda \rightarrow 1$ ) brings the fluctuation to zero, mass to zero, and charge to neutrality — the photon.

*f. The chain, and the plan of the paper.* Section II recalls the three postulates needed here. Section III formalizes charge as flow divergence. Section IV closes the conditional chain from accelerated charge to the photon: a time-varying source forces the field to vary; a varying field activates the nuclear channel; one flattening gives  $\mu = 1$ ; and  $\mu = 1$  is massless and neutral. Section V proves the exact sink-contraction theorems, and Sec. VI the half-phase theorems. Section VII applies the sink theorems to the two-flow-ring device. Section VIII reports the numerical witnesses; Sec. IX is the honest assessment — what is claimed, and what is not; Sec. X concludes. The chain is conditional by design: the paper proves that *if* a flattening occurs, any anchored substance becomes a photon; it does not prove that an accelerated charge *automatically* flattens its flow. That boundary, and the status of the input parameters  $\lambda, \delta, \varepsilon$ , are stated plainly in Sec. IX.

## II. POSTULATES OF THE FLOWING-SPACE FRAMEWORK

We recall the postulates needed here; full derivations are in Ref. [2]. Each postulate is stated, then read physically, so that the geometry of Secs. III–VII remains anchored.

**Postulate II.1** (Vector light speed). *Space itself flows at the vector speed  $C$  with  $|C| = c$ ; the direction of  $C$  is changeable.*

The river model of black holes [4] is the same picture: the medium moves, and what we call the speed of light is the speed of the medium. The direction being changeable is what allows flows to curl, converge, and diverge — the degrees of freedom used throughout this paper.

**Postulate II.2** (Three anchored directions). *Space has three independent orthogonal directions; a physical object is an anchored pattern of the flow, with anchor strength  $s \geq 0$ . The anchored mass is*

$$m_{\text{eff}}^2 = s^2(1 - \mu)^2, \quad (1)$$

where  $\mu \in [0, 1]$  is the antigravity strength (flattening of the flow gradient);  $\mu = 1$  gives  $m_{\text{eff}} = 0$  (*MassCancellation AMC1*).

Mass is anchoring: the more strongly a pattern grips the flow, the more massive it is — and the flatter the flow gradient, the less there is to grip. The quantity  $\mu$  measures how far the flattening has gone;  $\mu = 1$  is the completely flat flow, in which nothing can be anchored and everything is carried.

**Postulate II.3** (Four forces as Leibniz channels). *The flow momentum is  $P = m(C - v)$ ; its time derivative decomposes into four channels (GQF2):*

$$\frac{d}{dt}[m(C - v)] = \underbrace{\dot{m}C}_{\text{electric}} + \underbrace{m\dot{C}}_{\text{nuclear}} - \underbrace{\dot{m}v}_{\text{magnetic}} - \underbrace{m\dot{v}}_{\text{gravity}}. \quad (2)$$

Force is change of flow momentum, and the Leibniz rule splits that change into four named channels: change of mass while the medium flows (electric), change of the medium itself (nuclear), change of mass while the body moves (magnetic), and change of the body's own motion (gravity). The channel activated by a varying medium,  $m\dot{C}$ , is the one this paper will need.

### III. CHARGE AS FLOW DIVERGENCE

Charge is the divergence of the space-displacement flow (GQF4; VibrationChargeFlow CF1–CF5). The picture to hold on to is a whirlpool: a sink drains the surrounding surface toward itself, a source pushes it outward — the same medium, two opposite signs of convergence. This is also the oldest dynamical picture of the field itself: Maxwell built electrodynamics on the stresses of a medium [11], and here the medium is space.

**Theorem III.1** (Source and sink). *For a local divergence scalar  $\rho$ :*

$$\text{PositiveSource}(\rho) : \Longleftrightarrow 0 < \rho, \quad (3)$$

$$\text{NegativeSink}(\rho) : \Longleftrightarrow \rho < 0. \quad (4)$$

*Positive charge is a source (diverging displacement), negative charge a sink (converging displacement); both obey the right-handed helical motion of the flow, the sign of the divergence distinguishing them.*

The helical motion is shared — a source and a sink spiral the same way; only the sign of  $\rho$  tells them apart. Charge is not a new substance added to the flow; it is a local property of the flow itself, the divergence of its displacement.

**Theorem III.2** (Source–sink exclusivity; CF1–CF4). *(i) A source and a sink cannot hold simultaneously. (ii) Charge conjugation is the sign flip:  $0 < \rho \iff \rho < 0$  after  $\rho \mapsto -\rho$  (source and sink are exchanged). (iii) Zero divergence is neutral: neither source nor sink. (iv) The magnitude is preserved under the flip:  $|\rho| = |- \rho|$ .*

The four algebraic facts are the whole charge algebra in miniature: exclusivity (no point is both), conjugation (flipping the sign swaps the two), neutrality (zero divergence carries no charge), and magnitude (the amount of charge is untouched by conjugation). All four are theorems of the divergence structure, not assumptions about a quantum number.

The continuity side — the coupled extraction of the charge  $e$ , the divergence dynamics in 3D, and charge normalization — remains open (honestly registered in the repository). In particular, this paper does not yet derive the value of  $e$  or the discrete charge spectrum; it derives the geometric content of the sign.

#### IV. ACCELERATED CHARGE IMPLIES THE FIELD VARIES; THE NUCLEAR CHANNEL ACTIVATES

The story now turns from statics to dynamics. If charge is the divergence of the flow, an accelerated charge is a time-varying divergence: a source whose strength changes with time. What does that do to the space field  $C$ ? The sourced wave equation — the framework’s equation of motion for the medium, in the same family as light propagation through a dielectric [12] — answers in one stroke.

**Theorem IV.1** (CR1; time-varying source forces a varying field). *In the sourced wave equation  $\partial_t^2 C = c^2 \partial_x^2 C + J$  (MS5), a static field implies a static source: if  $\partial_t C \equiv 0$ , then  $J(t+1) = J(t)$  for all  $t$ . Hence a time-varying source (an accelerated charge) forbids a static space field.*

The contrapositive is the physical content: a field that does not change can only be fed by a source that does not change. Shake the charge, and the medium must shake with it — the field varies.

**Theorem IV.2** (CR2; the nuclear channel activates). *With  $m \neq 0$  and  $\dot{C} \neq 0$ , the nuclear channel of Eq. (2) is non-zero:  $m \dot{C} \neq 0$ .*

A varying medium means  $\dot{C} \neq 0$ , and any anchored body in it feels the nuclear channel of the Leibniz decomposition: the force that comes from the medium changing underneath the body, not from the body changing its own motion.

The missing link between the nuclear channel and the antigravity parameter  $\mu$  is not invented here: the framework already contains its own  $\mu$ -bridge. The bridge is the flattening progress of the flow — how much of the fluctuation energy has been removed — and it feeds  $\mu$  directly.

**Theorem IV.3** (CR3; one flattening gives  $\mu = 1$ ). *Let  $Q_A(v) = \sum_{i \in A} (v_i - \bar{v}_A)^2$  be the fluctuation energy, and  $\eta = 1 - Q_A(v)/Q_A(v_0)$  the flattening progress (MuFieldCoupling TD11). The  $\mu$  update is  $\mu' = \mu + \eta(1 - \mu)$  (PlasmaDynamics muStep). Flattening once gives  $\mu' = \mu + 1 \cdot (1 - \mu) = 1$ .*

The update rule says:  $\mu$  gains, per step, a fraction  $\eta$  of what remains between it and 1. A single complete flattening ( $\eta = 1$ ) therefore lands exactly on  $\mu = 1$  — not asymptotically, in one step.

**Theorem IV.4** (CR4;  $\mu = 1$  is the photon endpoint). *At  $\mu = 1$ , the anchored mass vanishes (Eq. (1) with  $1 - \mu = 0$ ) and the divergence is neutral (CF3). The object is massless and chargeless — the photon.*

Two conditions for a photon fall out of the same endpoint: the flat flow can hold no anchor (mass zero), and the flat flow has no divergence (charge zero). The photon of the heuristic quanta of 1905 [13] is here the completely unanchored, divergence-free pattern — the pattern carried entirely by the medium.

**Theorem IV.5** (CR5; closure of the chain). *One flattening applied to any region  $A$  (any anchored substance  $s$ ) yields a photon: mass zero and neutrality.*

The closure is universal: no property of the particular substance enters. Whatever is anchored, one flattening of its region transmutes it. The chain is conditional — accelerated charge  $\Rightarrow$  varying field  $\Rightarrow$  nuclear channel  $\Rightarrow$  flattening  $\Rightarrow \mu = 1 \Rightarrow$  photon — and its only unproved link is the second-input gap stated in Sec. IX: who applies the flattening.

## V. CONVERGENCE IS FLATTENING: SINK CONTRACTION

Why is the *negative* charge the natural antigravity carrier? The geometric answer: a sink is a converging flow, and convergence is flattening. Picture the whirlpool again: water that converges on the drain has, by the time it arrives, lost its individual differences — the surface is level at the center. The same contraction, applied to the space flow, removes the fluctuations that constitute internal structure. In miniature, this is what a warp contraction does to the space ahead of a traveler [14] — a contraction of the medium itself, not a motion through it; we use the picture only as an image.

**Theorem V.1** (Sink contraction; CR9). *For a region  $A$ , a field  $v : A \rightarrow \mathbb{R}$ , and  $\lambda \in [0, 1]$ ,*

$$\text{sinkContract}(A, v, \lambda)(i) = \begin{cases} (1 - \lambda) v_i + \lambda \bar{v}_A & i \in A, \\ v_i & i \notin A. \end{cases}$$

*Each point of  $A$  is pulled toward the region mean by a fraction  $\lambda$  — the discrete image of “the negative charge absorbs the surrounding flow toward itself.”*

The parameter  $\lambda$  is the strength of the sink per step:  $\lambda$  of the way toward the mean at every application. The operator is entirely local and needs no information beyond the current field — the sink acts by itself, which is why its effect will later be called self-flattening.

**Theorem V.2** (CR9; square decay of fluctuation). *The contraction preserves the region mean and scales the fluctuation energy by  $(1 - \lambda)^2$ :*

$$Q_A(\text{sinkContract}(A, v, \lambda)) = (1 - \lambda)^2 Q_A(v). \quad (5)$$

*For  $0 < \lambda < 1$  the decay is monotone and exact;  $\lambda = 1$  is a single flattening (fluctuation to zero). The sink is a self-flattening operator family.*

The fluctuation energy — the very quantity that measures how much internal structure the region carries — dies as  $(1 - \lambda)^2$ , quadratically, not linearly. Two facts make the theorem

exact rather than approximate: the mean is preserved (so the contraction has no secular drift), and the scaling is a square (so even a weak sink eventually empties the region of fluctuation).

**Theorem V.3** (CR10; strict attraction pointwise). *For  $0 < \lambda < 1$  and  $v_i \neq \bar{v}_A$ ,*

$$|\text{sinkContract}(A, v, \lambda)(i) - \bar{v}_A| < |v_i - \bar{v}_A|,$$

*each point is strictly attracted to the mean — “the sink pulls the surrounding flow to itself,” pointwise.*

Every point moves strictly closer on every step; nothing oscillates, nothing is left behind. The attraction is uniform in the sense that the contraction factor applies to every deviation, which is what makes the numerical approach (Sec. VIII) monotone.

The energy side of the sink is already present in the framework: information sinking into the absorbing branch  $iR$  releases  $\varphi(R) - \varphi(iR) = \delta + \varepsilon > 0$  (HiddenQFT HQ9a), so a converging flow carries its own flattening power as a candidate (CR8). This is a semantic correspondence (interface), not a theorem of the divergence dynamics — a distinction that Sec. IX keeps explicit.

## VI. THE HALF-PHASE GEOMETRY: SPIN 1/2 FROM THE CRITICAL SHEET

The second geometric addition concerns the value 1/2 itself. The guide is a logarithmic ruler: a full step multiplies by  $e$ ; the half step multiplies by  $\sqrt{e}$ . A mirror at the halfway point fixes the half: the reflection  $s \mapsto 1 - s$  keeps only 1/2 fixed, and in the logarithmic coordinate the mirror is the inversion  $w \mapsto e/w$ , whose fixed points lie on the circle  $|w| = \sqrt{e}$ . The identity  $\ln(\sqrt{e}) = 1/2$  is then not a curiosity of the logarithm but the statement that the fixed point of the reflection must sit at the halfway mark of the unit interval.

### A. The inversion fixed point

The Riemann–Hilbert envelope of the framework [10] carries the reflection  $s \mapsto 1 - s$ ; in the logarithmic coordinate  $w = e^s$  this is the inversion  $w \mapsto e/w$ . Its fixed set is the circle  $|w| = \sqrt{e}$  (envelope inversion fixed circle). Taking logarithms:

$$\ln(\sqrt{e}) = \frac{1}{2} \ln e = \frac{1}{2}. \tag{6}$$

**Theorem VI.1** (CS2; inversion fixed iff log half). *For  $r > 0$ ,*

$$r = e/r \iff \ln r = \frac{1}{2}.$$

*The fixed point of the reflection must sit at  $1/2$  in the logarithmic coordinate — because  $\ln e = 1$  and the fixed point of  $s \mapsto 1 - s$  is  $1/2$ .*

The theorem says the two facts are the same fact: being fixed under inversion (in the multiplicative picture) and being halfway (in the logarithmic picture). The value  $1/2$  is forced by the geometry of a reflection, not chosen.

### B. The half-angle phase is fermionic

The square root is the half-step operator of the multiplicative group; in the phase space of a vibration, the same half-step is the half-angle phase. Walk the unit circle in half steps: after one turn the pointer has advanced by only half a cycle — and a half cycle is a sign flip. The sign flip is the entire physical content of fermionic statistics at the level of phases.

**Theorem VI.2** (CS4; half-angle phase has closure factor  $-1$ ).  $\theta \mapsto \exp(i\theta/2)$  *satisfies*

$$f(\theta + 2\pi) = -1 \cdot f(\theta),$$

*i.e. it is a fermionic closure witness (VibrationStatistics ClosureFactor with  $\sigma = -1$ ):  $2\pi$  changes sign,  $4\pi$  restores. The half-angle phase is the explicit witness of spin  $1/2$  in the phase space of a vibration.*

This is the phase-geometric half of the spin–statistics story: a spinor needs  $4\pi$  to return to itself, and the half-angle phase is why. Dirac’s equation [8] encodes the same fact in  $\gamma$ -matrices; here it appears as the closure factor of the half-step map itself. Sec. IX states precisely what this theorem does and does not establish.

### C. Energy and mass

With  $E \propto m^2$  (mass–energy quadrature), logarithms are linear:  $\ln E = 2 \ln m$ . At the critical sheet,  $\ln(\sqrt{e}) = 1/2$  means  $m = \sqrt{e}$ ,  $E = e$ ; the factor 2 in  $\ln E = 2 \ln m$  is the inverse of the spin half-step — energy takes two half-steps to complete one cycle of  $m$ .

**Theorem VI.3** (CS5; logarithmic 2:1).  $\ln e = 2 \ln(\sqrt{e})$ .

The quadrature  $E \propto m^2$  and the half-step  $\ln(\sqrt{e}) = 1/2$  conspire: the factor 2 of the logarithm is the reciprocal of the half-step, so energy advances in mass at twice the rate that mass advances in itself — two half-steps per cycle, the fermionic signature read again in the energy–mass plane.

## VII. DEVICE APPLICATION: TWO-FLOW-RING OPTIMIZATION

The theorems of Sec. V are geometry; their purpose is engineering. The two-flow-ring device [15] drives the  $\mu$  dynamics of circulating electrons toward the working window  $\mu_{\text{work}} = 0.999$  with an external rotating magnetic field (RMF). The sink-flattening theorem (CR9) enters this  $\mu$  dynamics as a self-flattening gain:

$$\eta_{\text{sink}}(\lambda) = 2\lambda - \lambda^2, \quad (7)$$

the fraction of fluctuation removed by one contraction, added to the external RMF gain and decaying with the residual  $\eta_{\text{sink}} \cdot (1 - \mu)$  (the flattening loses its object as the region flattens; TD18). The comparison (evaluated numerically; honest candidate,  $\lambda$  an input):

TABLE I.  $\mu$  trajectory to the working window  $\mu_{\text{work}} = 0.999$  (FC11 ceiling 0.999780568).

|                                        | Pure external RMF | External + self-flattening ( $\lambda = 0.3$ ) |
|----------------------------------------|-------------------|------------------------------------------------|
| Steps to $\mu = 0.999$ (CR1 criterion) | 132               | 85                                             |
| Steps with halved external drive       | 267               | 150                                            |

The self-flattening shortens the approach to the working window by  $\sim 35\%$  and reaches it with halved external RMF drive — the sink’s own converging geometry contributes to the  $\mu$  production (supply-side, CR8 candidate). The negative result is registered honestly: a *constant*  $\eta_{\text{sink}}$  violates the FC11 ceiling (pushes  $\mu$  to 1); the decaying form is required. With a constant external drive,  $\mu$  still tends to 1 (a known TM-family behavior); stopping inside the window requires feedback shutoff (TD19), a separate matter.

## VIII. NUMERICAL VERIFICATION

All theorems of the chain are verified in Lean 4 with zero `sorry` (4348 jobs, full build green) [16] and numerically at machine precision where a numeric witness exists. The numerical checks reproduce known physics and known constants [17]; they are witnesses of the theorems, not new experimental predictions:

- 3D  $21^3$  lattice, negative-divergence field  $v = -r$ : every interior point has  $\text{div } v < 0$ ; initial fluctuation  $Q = 10187$ ; 200 sink-contraction steps drive  $Q \rightarrow 3.5 \times 10^{-61}$  with  $|\text{div}| \rightarrow 0$ , monotonically (`verify_sink_contract_3d`).
- Sink contraction scaling  $Q \rightarrow (1 - \lambda)^2 Q$  reproduced exactly (CR9 witness); strict pointwise attraction (CR10) monotone in  $\lambda$ .
- Numerical witness of the device comparison (`verify_sink_mu_optimization`):  $132 \rightarrow 85$  steps at equal external drive;  $267 \rightarrow 150$  with halved external drive;  $\eta_{\text{sink}}(\lambda)$  strictly increasing in  $\lambda$  (Tab. I).

## IX. HONEST ASSESSMENT

*a. Claimed.* (i) Charge is the divergence of the flow: source = positive divergence, sink = negative divergence, with the four algebraic facts (exclusivity, flip, neutrality, magnitude) formalized. (ii) The chain “accelerated charge  $\Rightarrow$  varying field  $\Rightarrow$  nuclear channel  $\Rightarrow$  one flattening  $\Rightarrow \mu = 1 \Rightarrow$  photon” is a closed conditional chain of theorems: *if* a flattening is applied, any anchored substance becomes a photon. (iii) A sink is geometrically a self-flattening operator: the exact discrete theorem  $Q_A \rightarrow (1 - \lambda)^2 Q_A$  under sink contraction. (iv) The half-step geometry locates spin  $1/2$ :  $\ln(\sqrt{e}) = 1/2$  is the fixed point of the envelope reflection in the logarithmic coordinate, and the half-angle phase is the explicit fermionic closure witness. (v) All statements are formalized in Lean 4 with zero `sorry`; the device comparison is numerically verified.

*b. Not claimed.* (i) The bridge “nuclear channel  $\Rightarrow$  flattening action” is *not* derived: who applies the flattening (the physical origin of the flattening power) is the second-input gap, unchanged. (ii)  $\lambda$  (the sink contraction rate) and  $\delta, \varepsilon$  (the sink energy scale) are inputs, not derived values;  $\eta_{\text{sink}} = 2\lambda - \lambda^2$  is a candidate implementation, not the unique mapping.

(iii) The continuum 3D correspondence “negative divergence  $\Rightarrow$  fluctuation functional decreases” is not formalized (the discrete lattice theorems are the algebraic seed). (iv) Spin  $1/2$  is *located* geometrically, not *derived* as a value: the other half of the spin–statistics theorem (phase  $\Rightarrow$  operator anticommutation) requires field-operator language — the language of quantum field theory and of fields on curved spacetime [18, 19] — absent from the framework; the result is half a theorem. (v) No new falsifiable prediction beyond the standard framework; all numerical values reproduce known physics.

The boundary is maintained throughout: what is claimed is certified (theorem plus proof, machine-checked); what is not yet a theorem is labeled as such in this section, never upgraded in the body.

## X. CONCLUSION

Two geometric structures extend the flowing-space framework. The divergence structure of charge makes the sink a self-flattening operator, closing — conditionally — the chain from accelerated charge to the photon and giving the two-flow-ring device a self-flattening gain that shortens the approach to the working window by a third. The half-phase geometry locates spin  $1/2$  at the envelope-reflection fixed point  $\ln(\sqrt{e}) = 1/2$ , with the half-angle phase as its explicit fermionic witness. Both additions are theorem-exact where they are exact, and honestly open where the inputs remain:  $\lambda$ ,  $\delta$ ,  $\varepsilon$ , and the continuum 3D correspondence.

The open items assemble into a short list of problems in the spirit of Hilbert’s program [20]: derive the flattening power from the divergence dynamics; derive  $\lambda$  and the sink energy scale; formalize the continuum 3D divergence-to-functional correspondence; and complete the spin–statistics theorem in field-operator language. Each is a well-posed question that this paper leaves with the boundary stated rather than blurred — the road is followed precisely, and the mountain is climbed only a little, just enough to be sure it is a mountain.

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